AI LAB 10.py - C:/Users/K Aishwarya/Desktop/AI/AI LAB 10.py (3.14.2)

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```
import heapq

graph = {
    'S': [('A', 1), ('B', 4)],
    'A': [('C', 2), ('D', 5)],
    'B': [('D', 1)],
    'C': [('G', 3)],
    'D': [('G', 3)],
    'G': []
}

heuristic = {
    'S': 7,
    'A': 6,
    'B': 4,
    'C': 3,
    'D': 2,
    'G': 0
}

def astar(start, goal):
    priority_queue = [(heuristic[start], 0, start, [start])]
    visited = set()

    print("A* SEARCH ALGORITHM")
    print("-------------------")
    print("Starting Node:", start)
    print("Goal Node:", goal)
    print("\nSearching for the shortest path...\n")

    while priority_queue:
        f, g, current, path = heapq.heappop(priority_queue)

        if current in visited:
            continue

        visited.add(current)

        print("Current Node:", current)
        print("Path:", " -> ".join(path))
        print("Path Cost (g):", g)
        print("Heuristic (h):", heuristic[current])
        print("Total Cost (f = g + h):", f)
        print()

        if current == goal:
            print("GOAL REACHED!")
```

```
==================================
A* SEARCH ALGORITHM
------------------
Starting Node: S
Goal Node: G

Searching for the shortest path...

Current Node: S
Path: S
Path Cost (g): 0
Heuristic (h): 7
Total Cost (f = g + h): 7

Current Node: A
Path: S -> A
Path Cost (g): 1
Heuristic (h): 6
Total Cost (f = g + h): 7

Current Node: C
Path: S -> A -> C
Path Cost (g): 3
Heuristic (h): 3
Total Cost (f = g + h): 6

Current Node: G
Path: S -> A -> C -> G
Path Cost (g): 6
Heuristic (h): 0
Total Cost (f = g + h): 6

GOAL REACHED!
Shortest Path: S -> A -> C -> G
Minimum Path Cost: 6
>
```